Lec 4. Perceptron & Generalized Linear model

Topic:

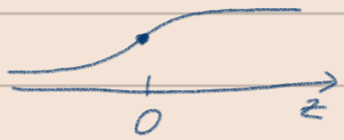
- perceptron Algorithm (historical but not used often)
- exponential family
- Generalized linear model
- Softmax regression (Multi class classification)

(판별 모델!)
4장까지는 Discriminative learning algorithm

$P(y|x)$ 주어진 입력에 대해 y가 나올 확률을 직접 모델링.

Perceptron algorithm 1950~60년대...

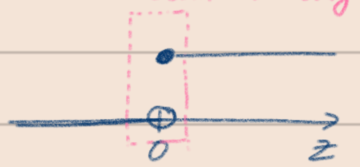
Logistic regression sigmoid



$$g(z) = \frac{1}{1+e^{-z}}$$

$$h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}$$

decision boundary



$$g(z) = \begin{cases} 1 & z \geq 0 \\ 0 & z < 0 \end{cases}$$ step function

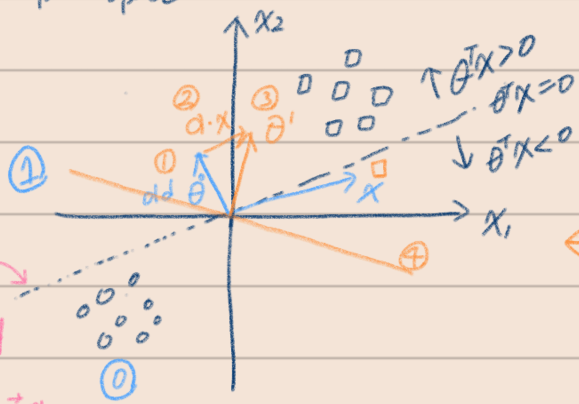
$$h_\theta(x) = g[\theta^T x]$$

$$\theta_j := \theta_j + \alpha \cdot \underbrace{(y^{(i)} - h_\theta(x^{(i)}))}_{} \cdot x_j^{(i)}$$

0: if the algorithm got it right
±1: 1 if wrong & $y^{(i)}=1$
−1 if wrong & $y^{(i)}=0$

$y^{(i)}$ 정답	$h_\theta(x^{(i)})$ 예측	차이	what to do?
1	1	0	No update
0	0	0	
1	0	1	예측이 낮음 → θ를 더 x쪽으로
0	1	−1	예측이 높음 → θ를 x와 멀어지게

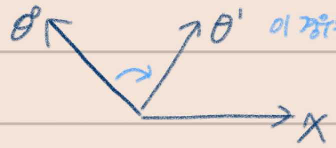
input space



decision boundary
$\theta^T x = 0$
→ 이때 $\vec{\theta}$는 $\vec{x}$에 orthogonal
& $\vec{\theta}$는 decision boundary에 대해 perpendicular

이 $\theta^T x = 0$을 돌려가며 binary classification
같은 그룹은 한 쪽에만 속하도록

where
$\theta \approx x \mid y=1$
$\theta \not\approx x \mid y=0$
we want θ to be similar to x in general
∵ when they are similar, the dot product > 0



이 경우는 ⊖

As θ is rotating, the decision boundary is kinda perpendicular to θ.
And you wanna get all the possible xs on one side of decision boundaries.

Exponential family a class of probablity distributions

whose PDF (probablity density function) can be written in the form: ⇐ 이게 정의 그자체.

$$P(y;\eta) = b(y)\exp[\eta^T T(y) - a(\eta)] = \frac{b(y)\exp(\eta^T T(y))}{e^{a(\eta)}}$$

형태만 맞으면 이산이든 연속이든 "지수분포군"

☆ y에 대하여 ☆ (θ에 대하여 아님) ⇓ θ에 대한것은 MLE ☆☆

y: data (we want to find it by the model) "scalar"

Eta (에타) η: natural parameter → 충분통계량과 '선형'으로 결합 (지수함수 안에서) parameter를 계산하기 좋게 바꾼것.

$T(y)$: sufficient statistic $= y$ (mostly y, no eta)

Dimension should match

$b(y)$: base measure (this function is of only y, cannot involve Eta η)

$a(\eta)$: log-partition function (only η & constants, no y) → η이 θ에 변형이나 a(η)의 gradient가 MLE에 사용됨 ☆☆

확률분포는 반드시 $\int p(x|\theta)dx = 1$ (분포의 총합이 1) 이어야 함

$a(\eta)$ 항은 전체 식의 적분이 1이 되도록 normalize 해줌. ('보정'의 역할)

Bernoulli (Binary data)

이산형 discret → PMF 특정 x값에 대해 $P(X=x)$ 계산 가능 (함수값 자체가 확률)

ϕ (phi '피') = probability of event (happening or not)

$p(y;\phi) = \phi^y(1-\phi)^{(1-y)}$ mathmatical way of representing if-else condition in progamming

↳ PMF of Bernoulli

Goal: massage PDF and see what a, b, T turn out to be

$$= \mathrm{Exp}(\log(\phi^y(1-\phi)^{(1-y)}))$$

$$= \underbrace{1}_{b(y)} \cdot \exp\left[\underbrace{\log\left(\frac{\phi}{1-\phi}\right)}_{\eta}\underbrace{y}_{T(y)} + \underbrace{\log(1-\phi)}_{a(\eta)}\right]$$

$$\eta = \log\left(\frac{\phi}{1-\phi}\right)y \Rightarrow \phi = \frac{1}{1+e^{-\eta}}$$

$$a(\eta) = -\log(1-\phi) \Rightarrow -\log\left(1-\frac{1}{1+e^{-\eta}}\right)$$

Gaussian (constant variance (fixed/known))

연속형 continuous → PDF 특정한 값에 대한 확률은 0. 구간에 대한 확률은 적분

$$P(a \le X \le b) = \int_a^b f(x)dx$$

Assume $\sigma^2 = 1$

$$P(y;\mu) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{(y-\mu)^2}{2}\right)$$

$$= \underbrace{\frac{1}{\sqrt{2\pi}}e^{-\frac{y^2}{2}}}_{b(y)}\exp(\underbrace{\mu}_{\eta}\underbrace{y}_{T(y)} - \underbrace{\tfrac{1}{2}\mu^2}_{a(\eta)})$$

강의노트

$b(y) = \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{y^2}{2}\right)$

$T(y) = y$

$\eta = \mu$

$a(\eta) = \frac{\mu^2}{2} = \frac{\eta^2}{2}$

<exponential familys>

Real – Gaussian

Binary – Bernoulli

count – Poisson

R+ – Gamma, Exponential dist

e.g. volume of an object, event time

P.Dist (over P.Dist) – Beta, Dirichlet

→ Bayesian

Nice mathmatical properties of Exponential families

1. MLE with respect to Eta η $\Rightarrow$ concave 오목함수 only global max

   NLL is convex negative log likelihood

no integrals needed

2. $E[y;\eta] = \frac{\partial}{\partial\eta} a(\eta)$ 확률분포를 정규화시켜주는 log partition fn $a(\eta)$의 gradient는 y의 기댓값 (예측하고자 하는 다음 레이블 y)
3. $Var[y;\eta] = \frac{\partial^2}{\partial\eta^2} a(\eta)$ ← 얘는 어디에? 두번 미분한 값은 y의 분산

GLM Generalized Linear Model

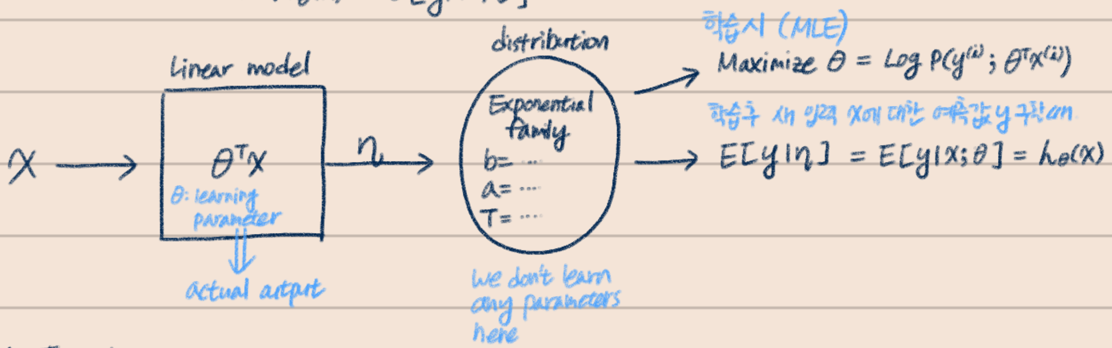
Assumptions / Design choices

(1) $y \mid x;\theta \sim$ Exponential family (η)

(2) $\eta = \theta^T x \qquad \theta \in \mathbb{R}^n,\ x \in \mathbb{R}^n$

(3) test time : output $E[y \mid x;\theta]$

$$\Rightarrow h_\theta(x) = E[y \mid x;\theta]$$



GLM Training

No matter which distribution, the learning update rule is the same

$$\theta_j := \theta_j + \alpha \cdot (y^{(i)} - h_\theta(x^{(i)}))\, x_j^{(i)}$$

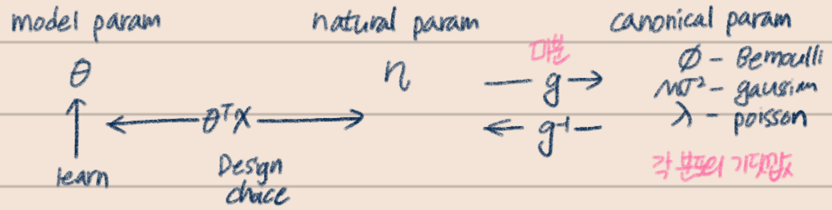
Terminology

η : natural parameter

$E[y;\eta] = g(\eta)$: canonical response function $\quad g(\eta) = \frac{\partial}{\partial\eta} a(\eta)$

$\eta = g^{-1}(\mu)$: canonical link function

3 - Parameterizations

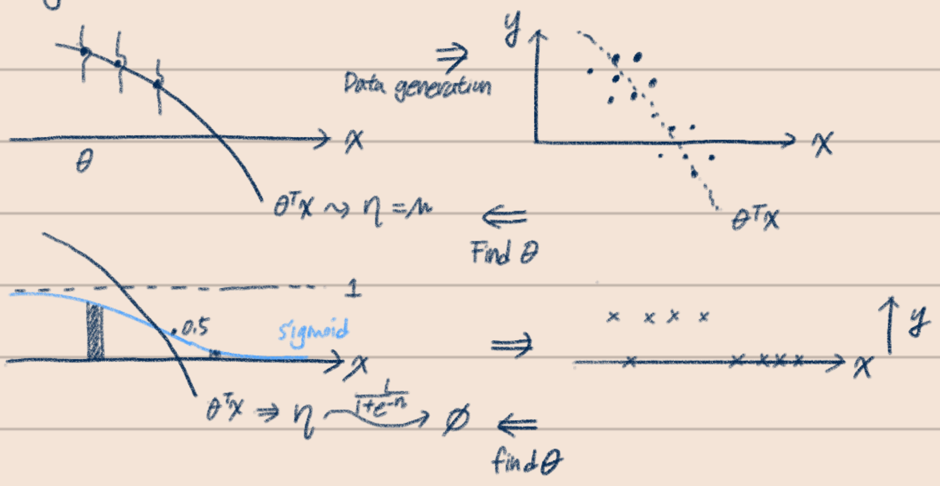


Logistic Regression → Bernoulli

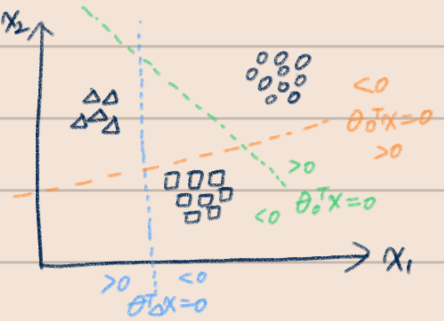
$h_\theta(x) = E[y \mid x;\theta] = \phi = \frac{1}{1+e^{-\eta}} = \frac{1}{1+e^{-\theta^T x}}$ Test 예측하고자 하는 y

Assumptions (visualized)

Regression



Softmax regression (cross entropy)

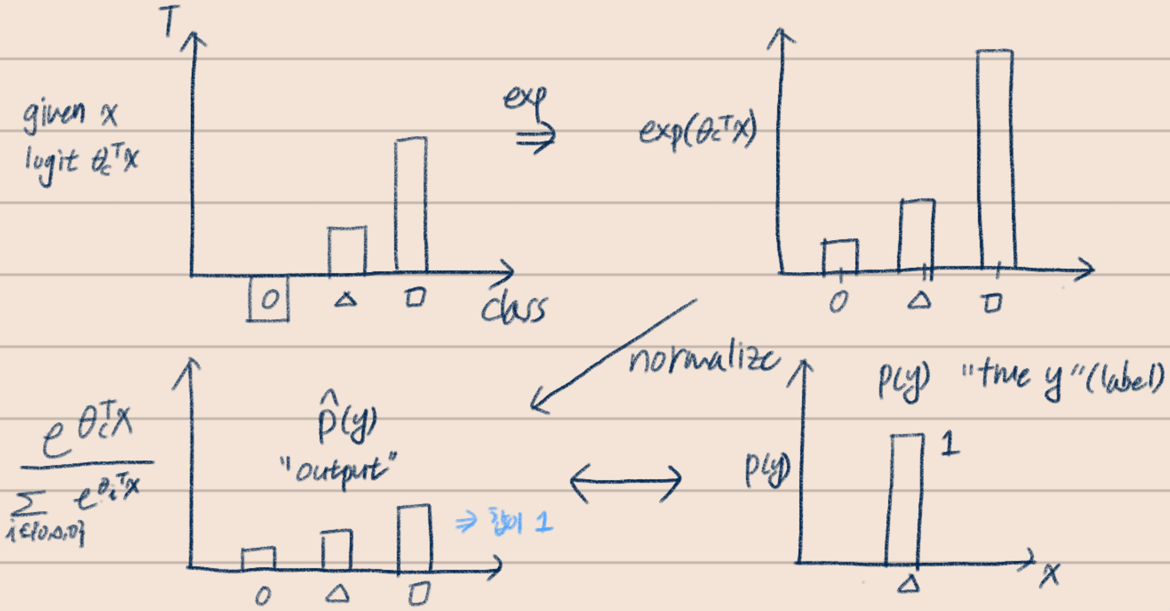


$x^{(i)} \in \mathbb{R}^n$

Label $y = [\{0,1\}^k]$ e.g. $[0,0,1,0]$

$\theta_{class} \in \mathbb{R}^n$ k such class $\in \{\Delta, O, \square\}$

$$k\left\{\begin{bmatrix} -\theta_0- \\ -\theta_1- \\ -\theta_{k-1}- \end{bmatrix}\right.$$
$\longleftarrow n \longrightarrow$



그동안은 output이
scalar 또는 probality 였지만
$\hat{p}(y)$은 여러 클래스에 확률 분포

Δ class에만 1 ∴ 나머지항 소거됨

$$\text{cross entropy}(p, \hat{p}) = -\sum_{y \in \{O,\Delta,\square\}} p(y) \log \hat{p}(y)$$

를 줄이다.

$$= -\log \hat{p}(y_\Delta)$$

$$= -\log \frac{e^{\theta_\Delta^T x}}{\sum_{c \in \{\Delta O \square\}} e^{\theta_c^T x}}$$

⇑

treat as loss, do gradient descent